

1. What is DriverManager.getConnection(url, user, pass)?

It is used to **create a connection** between a Java application and the database.

- url → database URL
- user → database username
- pass → database password

It returns a **Connection object**, which is used to execute SQL queries.

2. What is createStatement() in JDBC?

createStatement() is a method of the **Connection interface**.

- It is used to create a **Statement object**
- Used when SQL queries **do not have parameters**

Common methods of Statement:

- executeQuery() → for SELECT
- executeUpdate() → for INSERT, UPDATE, DELETE
- execute() → for any SQL

3. What is PreparedStatement in JDBC?

PreparedStatement is used to execute **parameterized SQL queries**.

- It is created using connection.prepareStatement(sql)
- It is **more secure and faster** than Statement
- Prevents **SQL injection**

Common methods:

- setInt(), setString(), setLong()
- executeQuery()
- executeUpdate()

4. Difference between createStatement() and PreparedStatement

Statement	PreparedStatement
No parameters	Supports parameters
Less secure	Prevents SQL injection
Slower	Faster
SQL written every time	SQL compiled once

5. JDBC execution flow using all components

1. Load JDBC driver (**JDBC 4.0 onward**, drivers are **auto-loaded** via Service Provider mechanism.
2. Create connection using DriverManager.getConnection()
3. Create Statement or PreparedStatement
4. Execute SQL query
5. Get result using ResultSet
6. Close all resources

6. Why is PreparedStatement preferred?

- Prevents **SQL injection**
- Improves **performance**
- Code is **cleaner and reusable**
- Handles parameters safely

7. Difference between executeQuery(), executeUpdate(), and execute()

- executeQuery() → SELECT → returns ResultSet
- executeUpdate() → INSERT, UPDATE, DELETE → returns number of rows
- execute() → used when query type is unknown

8. How does JDBC handle transactions?

By default, JDBC uses **auto-commit mode**.

- `setAutoCommit(false)` → start manual transaction
- `commit()` → save changes
- `rollback()` → undo changes if error occurs

Transactions ensure **atomicity, consistency, isolation, and durability (ACID)** in database operations.

9. What is DataSource and how is it different from DriverManager?

DataSource is used to get database connections **with connection pooling**.

DriverManager	DataSource
No pooling	Supports pooling
Slower	Faster
Manual management	Managed by container

10. What is connection pooling?

Connection pooling means:

- Reusing database connections
- Avoiding creation of new connections every time

This improves:

- Performance
- Scalability
- Resource usage

11. How do you close JDBC resources?

Always close:

1. `ResultSet`
2. `Statement / PreparedStatement`

3. Connection

Best way:

- Use **try-with-resources**
- Prevents memory leaks

12. What is UserDetailsService and JdbcUserDetailsManager?

UserDetailsService is a Spring Security interface used to load user information during authentication. It provides the `loadUserByUsername()` method, which returns user details like username, password, and roles. JdbcUserDetailsManager is a built-in implementation of UserDetailsService that fetches user details from a database using JDBC and a DataSource, without writing custom authentication logic.